

MYSTERY PACKAGING

DESCRIPTION : At the beginning of the event, teams will be given a bag of building materials and instructions for designing a package that will protect a given material from a drop. This is an ON-SITE BUILD EVENT.

COMPETITORS BRING/NEED : (per student)

- Allowable tools (Writing utensil, Scissors, Ruler)
- Safety goggles

NUMBER OF PARTICIPANTS : Maximum of 2

IMPOUND : NO

THE COMPETITION :

BUILD:

1. All teams will build at the same time. Teams will have 25-minutes to build. Once teams enter the area to build, they may not leave the area or receive outside assistance, materials or communication until after they are finished building. Violation of this rule places a team in Tier 2.
2. Once participants arrive at their build area, they must put on their safety goggles. These goggles must stay on for the entire build period. Supervisors will give one goggle warning. Additional warnings will place the team in Tier 2 and/or remove them from competition.
3. Each team will be given a bag containing the same materials and working area.
 - a. Examples of materials are : paper cups, drinking straws, paper clips, string, tape, paper, Play Doh, and craft sticks. Materials are not limited to this list but will not include anything potentially hazardous.
 - b. Teams may not use any outside materials, other than what is provided to them by the event supervisor or their allowable materials listed above.
 - c. The bag that the materials come in cannot be part of the package design and should be used to hold all waste at the end of the build.
 - d. The materials provided may be altered as needed, using the allowable tools.
 - e. The allowable tools may not be part of the package.
4. The material that will be loaded into the device will be announced at the start of construction.
 - a. This could be either a solid or a liquid and either homogeneous or heterogeneous in composition.
 - b. After the teams have constructed the package, the material will be loaded into the device during the test phase.
 - c. Teams will be allowed to examine a separate "inspection" load material of the same size and weight as used for testing but this cannot be loaded into the device during construction. This inspection area will be separate from the building area and may be accessed by any team, however teams may not take the sample material back to their building area.
5. The team will create a package using the given materials that will hold the load when it is dropped from a height between 0.5 and 1 meter, depending on the team's choice.
 - a. The device must be built to allow for loading of the material after construction has been completed.
 - b. The loading of the material should be able to be completed and the device ready to drop within one minute.
 - c. The package must have the team number on it. Packages without team numbers will be placed in Tier 2.
6. At the end of the build time, teams must clean up their build area by placing all trash and extra materials in the bag.
 - a. The remaining supplies will be massed and this will be used as a tie-breaker (those with more supplies left over will place higher in the tie.)
 - b. Prior to leaving the build area, teams must ensure that their area is clean and all instructions have been followed. Failure to follow this rule will place teams in Tier 2.

- c. Teams should store their packages in a location noted by the event supervisor. It is suggested that the event supervisor/judges do not touch student packages and instead have them place the packages in a location that will be monitored throughout the day.

TEST:

1. Teams will return to test at some point later in the day - this may be self-scheduled or based on a set schedule.
2. Once participants arrive at their test area, supervisors will have them collect their device and give them the loading material (this will be the same for all participants.)
3. Teams have one minute to load the material into their package and balance the package on the scale to measure the mass. Teams who do not meet this time will earn participation points only.
4. Teams whose package cannot hold the entire load will earn participation points only.
5. Teams will then take their full package and proceed to the testing site.
6. One member of the team will hold a meter stick and the other will hold the meterstick and the other will hold the package at the drop height of their choice. This drop height should be between 0.5 and 1 meter. The height is measured from the bottom of the package to the floor. The supervisor will confirm the height.
7. One given the signal by the event supervisor, teams will drop their full package. Teams have one chance to drop their package. There are no trial runs or do-overs.
8. Teams will pick up their package and carry it to the scale to measure the ending mass to account for lost load. Any material that falls out prior to massing will not count in the final mass.
9. Teams will remove the load according to the supervisor's instructions and return their package to the location where they were stored. Teams may pick up their packages at the end of the day, if desired.

RELATED FLORIDA STANDARDS:

SC.5.N.1.5 - Recognize and explain that authentic scientific investigation frequently does not parallel the steps of "the scientific method"

SC.5.P.13.1 - Identify familiar forces that cause objects to move, such as pushes or pulls, including gravity acting on falling objects.

SCORING:

1. High Score wins.
2. The following formula will be used to get the test score:
$$\text{Score} = [\text{drop height} - (\text{initial loaded package mass} - \text{final package mass})] \times 100$$
 - Drop height will be measured in centimeters
 - Mass will be measured in grams
 - Negative point values are allowed
3. Devices that cannot support the load will be given participation points only.

Team# _____

School Name: _____

Mystery Packaging Scoring Sheet

BUILD:		TIER
Outside assistance violation?	Y= Tier 2; Repeated violations may result in DQ	
Safety goggles violation?	1 st time= Warning; 2 nd time = Tier 2; Repeated violations may result in DQ	
Device made only of allowable materials?	N= Participation points only	
Device has team number on it?	N= Tier 2	
Building area cleaned up as instructed?	N = Tier 2	
Team did not show up at build time?	Y= Participation points only	
TIER?		

TEST:

Mass of bag with "waste"	_____ g (TIE-BREAKER)	
Loaded within 1-minute?	N= Participation points only	
Device held the load?	N= Participation points only	
Mass of fully loaded package BEFORE drop in grams		
Height of drop in centimeters		
Mass of package and remaining load AFTER drop in grams		
Change in Mass		

Drop Height in cm	-	Change in mass in grams	X	100 =	Final Score
	-		X	100 =	

PLEASE NOTE THAT THIS SCORESHEET IS A SAMPLE AND GUIDE- IN CASES OF CONFLICT WITH THE RULES,
THE RULES TAKE PRECEDENCE OVER THE SCORESHEET.